

Experiment No. 6

DNS Packet Analysis (Windows only)

Analysis of Application Layer Packets (DNS) using DNS commands & Wireshark in Windows

Objective:

- To understand the working of DNS as an application layer protocol.
- To analyze DNS packets generated by DNS commands.
- To capture and study DNS packets using Wireshark on Windows.

Software/Tools Required:

- Windows PC with Internet access
- Command Prompt (`cmd`)
- Wireshark (download from <https://www.wireshark.org/>)

Step-by-step Procedure:

Part A: Using DNS Commands (`nslookup`)

Step 1: Open Command Prompt

- Press `Win + R`, type `cmd`, and press `Enter`.

Step 2: Find IP address of a domain

```
nslookup www.example.com
```

- Observe:
 - DNS server used (often your ISP's DNS or local DNS server)
 - The resolved IP address

Step 3: Perform a reverse lookup (IP → domain)

```
nslookup 93.184.216.34
```

- See the canonical domain name corresponding to the IP.

Step 4: Query different DNS record types (e.g., mail servers)

```
nslookup -query=MX example.com
```

- Shows mail exchange servers for the domain.

Part B: DNS Packet Analysis with Wireshark

Step 1: Install and open Wireshark

- Download and install Wireshark.
- Open Wireshark as Administrator.

Step 2: Start capturing packets

- In the Wireshark main window, select your active interface (usually `Wi-Fi` or `Ethernet`).
- Click the **blue shark fin** icon to start capture.

Step 3: Generate DNS traffic

- In Command Prompt, run:

```
nslookup www.example.com
```

Step 4: Stop capturing

- Click the **red square** (Stop) in Wireshark.

Step 5: Filter DNS packets

- In the filter bar at the top of Wireshark, type:

```
dns
```

- Press `Enter` to filter and display only DNS-related packets.

Step 6: Analyze captured packets

- Select the DNS query packet (from your PC to DNS server, usually UDP port 53).

- Expand:
 - **Transaction ID:** used to match request & response
 - **Flags:** standard query, recursion desired, response
 - **Questions:** the domain name queried
- Select the response packet from DNS server:
 - Observe the resolved IP address under the **Answers** section.

Observation Table

Step	Command / Action	Observation
1	nslookup <u>www.example.com</u>	Resolved IP address shown
2	nslookup -query=MX example.com	Mail server records shown
3	Wireshark capture	DNS query and response packets
4	Packet details	Transaction ID, flags, domain name, IP address

Result:

- Successfully generated DNS requests using `nslookup`.
- Captured and analyzed DNS packets using Wireshark on Windows.

Conclusion:

- DNS translates domain names to IP addresses.
- Using Wireshark on Windows helps visualize real DNS query-response exchanges.

Output / Screenshots

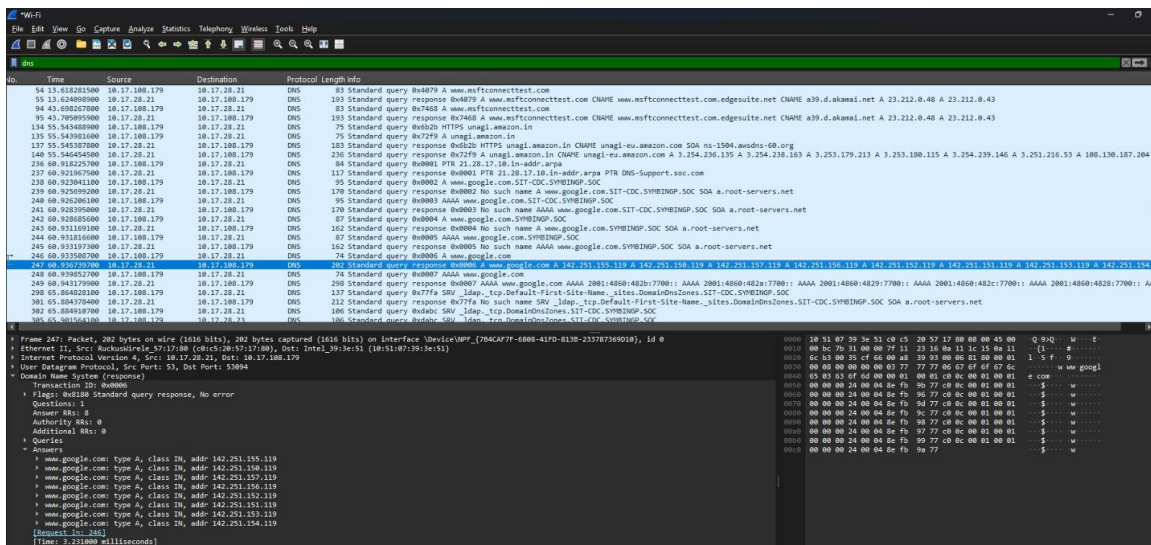


Fig 1

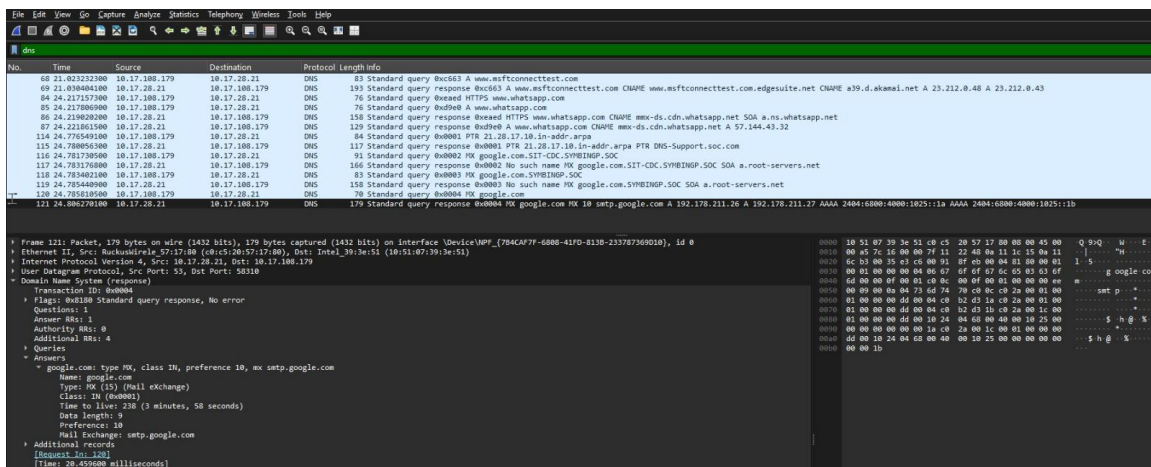


Fig 2

```
C:\Users\sit.lab2>nslookup www.google.com
Server:  DNS-Support.soc.com
Address:  10.17.28.21

Non-authoritative answer:
Name:     www.google.com
Addresses: 2001:4860:482b:7700::
           2001:4860:482a:7700::
           2001:4860:4829:7700::
           2001:4860:482c:7700::
           2001:4860:4828:7700::
           2001:4860:4826:7700::
           2001:4860:4827:7700::
           2001:4860:482d:7700::
           142.251.155.119
           142.251.150.119
           142.251.157.119
           142.251.156.119
           142.251.152.119
           142.251.151.119
           142.251.153.119
           142.251.154.119

C:\Users\sit.lab2>nslookup 142.251.155.119
DNS request timed out.
    timeout was 2 seconds.
Server:  UnKnown
Address:  10.17.28.21

*** UnKnown can't find 142.251.155.119: Non-existent domain

C:\Users\sit.lab2>nslookup -query=mx google.com
Server:  DNS-Support.soc.com
Address:  10.17.28.21

Non-authoritative answer:
google.com      MX preference = 10, mail exchanger = smtp.google.com

smtp.google.com internet address = 192.178.211.27
smtp.google.com internet address = 192.178.211.26
smtp.google.com AAAA IPv6 address = 2404:6800:4000:1025::1b
smtp.google.com AAAA IPv6 address = 2404:6800:4000:1025::1a
```

Fig 3